

ILLINOIS TECH | College of Computing

CH22: Ethical Aspects of Emerging and Converging Technologies

P2: Bioinformatics, Nanotechnology and Nanocomputing



27

2. Nanotechnology and Nanocomputing

- A number of ethical and social controversies arise at the intersection of two distinct technologies that are now also converging – viz., cybertechnology and nanotechnology.
- Ruth Chadwick and Antonio Marturano (2006) argue that nanotechnology provides the “key” to technological convergence in the 21st century

29

Genomics data reaches unprecedented scale

The growing reliance on AI in discovery is closely linked to the scale of biological data now being generated, particularly in genomics. Advances in sequencing technologies continue to accelerate data generation, but analysis remains a major bottleneck.



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“In 2026, we will see stunning examples of how AI can interpret genomics data to understand complex biology,” says Neil Ward, Vice President and General Manager for EMEA at PacBio. He highlights the scale of the challenge, noting that analysing a single genome involves tens of thousands of lines of code. At population scale, genomics studies “could generate up to 15x more data than YouTube over the next decade.”

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11-1-6

31

Module Objectives

Upon completion of this lesson the students should be able to:

- Define **nanotechnology** and describe the ethical concerns associated with it
- Describe and discuss ethical concerns affecting **bioinformatics** research and development

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Slide 1-28

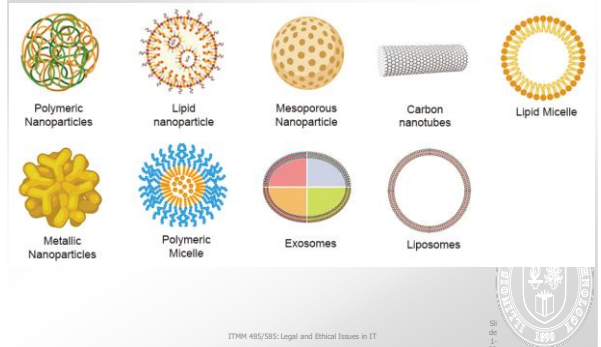
28

Defining Nanotechnology

- Rosalind Berne (2015) defines **nanotechnology** as “the study, design, and manipulation of natural phenomena, artificial phenomena, and technological phenomena at the nanometer level.”
- K. Eric Drexler, who (many believe) coined the term **nanotechnology** in the 1980s, describes the field as a **branch of engineering dedicated to the development of extremely small electronic circuits and mechanical devices built at the molecular level of matter.**

30

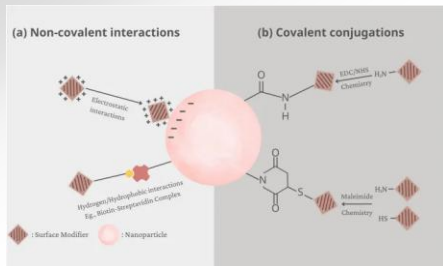
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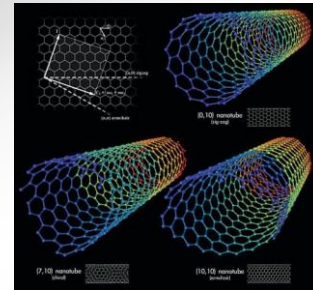
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32



<https://www.frontiersin.org/journals/nanotechnology/articles/10.3389/fnano.2024.1456939/full>

33



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34

Nanotechnology and Nanocomputing

- Drexler (1991) predicted that developments in nanotechnology will result in computers at the nano-scale, no bigger in size than bacteria, called *nanocomputers*.
- Nanocomputers can be designed using various types of architectures.
- An electronic nanocomputer would operate in a manner similar to present-day computers, differing primarily in terms of size and scale.

35

Nanoethics: Identifying and Analyzing Ethical Issues in Nanotechnology

- Moor and Weckert (2004) believe that assessing ethical issues that arise at the nano-scale is important because of the kinds of “policy vacuums” that are raised.
- They do not argue that a separate field of applied ethics called *nanoethics* is necessary.
- But they make a strong case for why an analysis of ethical issues at the nano-level is now critical.

36

Nanoethics (Continued)

- Moor and Weckert identify three distinct kinds of ethical concerns at the nano-level that warrant analysis:
 1. **privacy and control;**
 2. **longevity;**
 3. **runaway nanobots.**

37

Ethical Aspects of Nanotechnology: Privacy Issues

- Moor and Weckert note that we will be able to construct nano-scale information-gathering systems that can also track people.
- It will become extremely easy to put a nano-scale transmitter in a room, or onto someone’s clothing.
- Individuals may have no idea that these devices are present or that they are being monitored and tracked by them.

38



Ethical Aspects of Nanotechnology: Longevity Issues

- Moor and Weckert note that while many see longevity as a good thing, there could also be negative consequences.
- For example, they point out that we could have a population problem if the life expectancy of individuals were to change dramatically.

39



Ethical Aspects of Nanotechnology: Longevity Issues (Continued)

- Moor and Weckert also point out that if fewer children are born relative to adults, there could be a concern about the lack of new ideas and “new blood.”
- The authors also note that questions could arise with regard to how many “family sets” couples, whose lives could be extended significantly, would be allowed to have during their expanded lifetime.

40



Ethical Aspects of Nanotechnology: Runaway Nanobots

- Moor and Weckert further point out that when nanobots work to our benefit, they build what we desire.
- But when nanobots work incorrectly, they can build what we don't want.
- Some critics worry that the (unintended) replication of these bots could get out of hand.
- Many now refer to the replication problem (of nanobots) as the “grey goo scenario.”

41



Should Research/Development in Nanocomputing Be Allowed to Continue?

- Joseph Weizenbaum (1976) argued that computer science research that can have “irreversible and not entirely unforeseeable side effects” should not be undertaken.
- Bill Joy (2000) has argued that because developments in nanocomputing are threatening to make us an “endangered species,” the only realistic alternative is to limit its development.
- If Joy and others are correct about the dangers of nanotechnology, we must seriously consider whether research in this area should be limited.

42



Should Nanotechnology Research/ Development Be Prohibited?

- Ralph Merkle (2001) would disagree with Joy and others on limiting nano-level research.
- Merkle argues that if research in nanotechnology is prohibited, or even restricted, it will be done “underground.”
- If this happens, nano research would not be regulated by governments and by professional agencies concerned with social responsibility.

43



Should We Presume in Favor of Continued Nano Research?

- Weckert (2006) argues that potential disadvantages that *could* result from research in a particular field are not in themselves sufficient grounds for halting research.
- He suggests that there should be a *presumption in favor of freedom in research*.
- But Weckert also argues that it should be permissible to restrict or even forbid research where it can be clearly shown that harm is *more likely than not* to result from that research.

44



Assessing Nanotechnology Risks: Applying the Precautionary Principle

- Questions about how best to proceed in scientific research when there are concerns about harm to the public good are often examined via the *Precautionary Principle*.
- Weckert and Moor (2004) interpret the precautionary principle to mean the following:
If some action has a possibility of causing harm, then that action should not be undertaken or some measure should be put in its place to minimize or eliminate the potential harms.

45



Nanotechnology, Risk, and the Precautionary Principle (Continued)

- Weckert and Moor believe that when the precautionary principle is applied to questions about nanotechnology research and development, it needs to be analyzed in terms of three different “categories of harm”:
 - 1) *direct harm*,
 - 2) *harm by misuse*,
 - 3) *harm by mistake or accident*.
- The kinds of risks for each differ significantly.

46



Nanotechnology, Risk, and the Precautionary Principle (Continued)

- With respect to *direct harm*, Weckert and Moor analyze a scenario in which the use of nanoparticles in products could be damaging to the health of some people.
- They also note that the kinds of risks involved in direct harm are very different from those arising in the example they use to illustrate *harm by misuse* – i.e., developments in nano-electronics that could endanger personal privacy.
- Regarding *harm by mistake or accident*, Weckert and Moor describe a scenario in which nanotechnology could lead to the development of self-replicating, and thus “runaway,” nanobots. (This kind of harm will occur *only if* mistakes are made or accidents occur).

47



Nanotechnology, Risk, and the Precautionary Principle (Continued)

- Weckert and Moor argue that when assessing the risks of nanotechnology via the precautionary principle, we need to look at not only potential harms per se, but also at the *relationship* between “the initial action and the potential harm.”
- In their example involving direct harm, the relationship is fairly clear and straightforward: we simply need to know more about the scientific evidence for nanoparticles causing harm.
- In their case involving potential misuse of nanotechnology, e.g., in endangering personal privacy, the relationship is less clear.
- In the case of the third kind of harm, Weckert and Moor claim that we need evidence regarding the “propensity of humans to make mistakes or the propensity of accidents to happen.”

48



Nanotechnology, Risk, and the Precautionary Principle (Continued)

- Weckert offers the following solution or strategy:
If a *prima facie* case can be made that some research will likely cause harm...then the burden of proof should be on those who want the research carried out to show that it is safe.
- He also believes that there should be:
...a presumption in favour of freedom until such time a *prima facie* case is made that the research is dangerous. The burden of proof then shifts from those opposing the research to those supporting it. At that stage the research should not begin or be continued until a good case can be made that it is safe.

49